Lecture 15, Math 441

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Derivation of one of Euler's formulas

Consider the IVP: $y'' + y = 0; \qquad y(0) = 1; \;\; y'(0) = 0$

Using $y = \mathrm{e}^{rx}$ we get the characteristic equation $r^2 + 1 = 0 \to r = \pm \mathrm{i}$

and the general solution $y(x) = c_1\,\mathrm{e}^{\mathrm{i}x} + c_2\,\mathrm{e}^{-\mathrm{i}x}$

$y' = \mathrm{i}c_1\,\mathrm{e}^{\mathrm{i}x} - \mathrm{i}c_2\,\mathrm{e}^{-\mathrm{i}x}$

Apply the initial conditions:

$y(0) = c_1 + c_2 = 1 \qquad y'(0) = \mathrm{i}c_1 - \mathrm{i}c_2 = 0 \qquad \Longleftrightarrow \; c_1 = c_2 = \frac{1}{2}$

$$\boxed{y = \frac{1}{2}\,\mathrm{e}^{\mathrm{i}x} + \frac{1}{2}\,\mathrm{e}^{-\mathrm{i}x}}$$

(the solution to the IVP)

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We know that another general solution is

$y(x) = A\cos(x) + B\sin(x)$ (you can easily verify this)

$y'(x) = -A\sin(x) + B\cos(x)$

Apply the initial conditions: $y(0) = A = 1$ and $y'(0) = B = 0$

So the solution to the IVP is $y = \cos(x)$

$\cos(x) = \frac{1}{2}\left(e^{ix} + e^{-ix}\right)$

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Constant Coefficient Equations (cont.)

Ex: $y'' + 6\,y' + 13\,y = 0$ Use $y = \mathrm{e}^{rx}$ to get the characteristic

equation $r^2 + 6r + 13 = 0$

$r = -3 \pm \sqrt{9 - 13} = -3 \pm 2\mathrm{i}$

$y_1 = \mathrm{e}^{(-3+2\mathrm{i})x}$ $\qquad$ $y_2 = \mathrm{e}^{(-3-2\mathrm{i})x}$

General solution: $y(x) = c_1\,\mathrm{e}^{(-3+2\mathrm{i})x} + c_2\,\mathrm{e}^{(-3-2\mathrm{i})x}$

We can rewrite this as

$$y(x) = \mathrm{e}^{-3x}\left[c_1\mathrm{e}^{2\mathrm{i}x} + c_2\mathrm{e}^{-2\mathrm{i}x}\right] = \mathrm{e}^{-3x}\left[A\cos(2x) + B\sin(2x)\right]$$

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Higher Order Linear Homogeneous ODE

The standard notation is (n_{th} order):

$$y^{(n)} + p_1(x)\, y^{(n-1)} + p_2(x)\, y^{(n-2)} + \cdots + p_{(n-1)}(x)\, y' + p_n(x)\, y = 0$$

Principle of superposition: If $y_1, y_2, \cdots, y_k$ are solutions of the linear homogeneous equation, then any linear combination

$y = c_1\, y_1 + c_2\, y_2 + \cdots + c_k\, y_k$ is also a solution.

Linear Independence

Given n solutions $y_1(x), y_2(x), \cdots, y_n(x)$, they are

linearly independent if none is a linear combination of the others.

Ex: 1, e^{x}, e^{-x} are linearly independent:

$1 \neq a_1 \mathrm{e}^{x} + a_2 \mathrm{e}^{-x}$ $\quad$ $\mathrm{e}^{x} \neq a_1 1 + a_2 \mathrm{e}^{-x}$ $\quad$ $\mathrm{e}^{-x} \neq a_1 1 + a_2 \mathrm{e}^{x}$

Ex: $\cos(x)$, $\mathrm{e}^{\mathrm{i}x}$, $\mathrm{e}^{-\mathrm{i}x}$ are linearly dependent:

$\cos(x) = \frac{1}{2}\mathrm{e}^{\mathrm{i}x} + \frac{1}{2}\mathrm{e}^{-\mathrm{i}x}$

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Ex: 1, x, x^2 are linearly independent. $\cos(2x)$, $\cos(x)$, $\sin(x)$ are linearly independent. 3, x, $2x+4$ are linearly dependent.

For $\cos(x)$, $\mathrm{e}^{\mathrm{i}x}$, $\mathrm{e}^{-\mathrm{i}x}$, which are linearly dependent, we can write:

$\cos(x) - \frac{1}{2}\mathrm{e}^{\mathrm{i}x} - \frac{1}{2}\mathrm{e}^{-\mathrm{i}x} = 0$

or: $2\cos(x) - \mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x} = 0$

or: $-\cos(x) + \frac{1}{2}\mathrm{e}^{\mathrm{i}x} + \frac{1}{2}\mathrm{e}^{-\mathrm{i}x} = 0$

and infinite other combinations.

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$a_1 \cos(x) + a_2 \mathrm{e}^{\mathrm{i}x} + a_3 \mathrm{e}^{-\mathrm{i}x} = 0$

there are infinite choices for a_1, a_2, a_3 to satisfy the equality.

For $a_1 1 + a_2 \mathrm{e}^{x} + a_3 \mathrm{e}^{-x} = 0$, which are linearly independent, there is

only one choice for a_1, a_2, a_3, namely, $a_1 = a_2 = a_3 = 0$.

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An Equivalent Definition of Linear Independence

Given n solutions $y_1(x), y_2(x), \cdots, y_n(x)$, write the equation

$a_1\, y_1(x) + a_2\, y_2(x) + \cdots + a_n\, y_n(x) = 0$

If $a_1 = a_2 = \cdots = a_n = 0$ is the only choice for the equality to hold, then $y_1(x), y_2(x), \cdots, y_n(x)$ are linearly independent (LI). If there are other (infinite) choices, then $y_1, \cdots, y_n$ are linearly dependent (LD).

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$$a_1\, y_1(x) + a_2\, y_2(x) + a_3\, y_3(x) = 0$$

Take two derivatives:

$$a_1\, y_1'(x) + a_2\, y_2'(x) + a_3\, y_3'(x) = 0$$

$$a_1\, y_1''(x) + a_2\, y_2''(x) + a_3\, y_3''(x) = 0$$

$$W(y_1, y_2, y_3)(x) = \det \begin{pmatrix} y_1(x) & y_2(x) & y_3(x) \\ y_1'(x) & y_2'(x) & y_3'(x) \\ y_1''(x) & y_2''(x) & y_3''(x) \end{pmatrix}$$

$$\begin{cases} \neq 0 & a_1 = a_2 = a_3 = 0 \text{ is the only possibility } \Longrightarrow y_1,\, y_2,\, y_3 \text{ LI.} \\ = 0 & \infty \text{ solutions for } a_1,\, a_2,\, a_3 \Longrightarrow y_1,\, y_2,\, y_3 \text{ LD.} \end{cases}$$

If $y_1, \cdots, y_n$ are solutions in an interval I, then ...

- $y_1, \cdots, y_n$ are LI $\implies$ $W(y_1, \cdots, y_n)(x) \neq 0 \ \forall x \in I$.
- $y_1, \cdots, y_n$ are LD $\implies$ $W(y_1, \cdots, y_n)(x) = 0 \ \forall x \in I$.

Remember:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$